Predictors of Earnings Risk with Machine Learning

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Abstract

This paper looks at the predictors of lifetime earnings risk under a Restricted Income Profile (RIP) model using traditional and machine learning methods such as lasso and SHAP values. The paper builds on the work of Drewianka and Oberg (2025) which uses a moment condition approach derive a parameter that captures permanent income risk. The paper finds that education and age are important in explaining lifetime earnings risk. The paper also finds that macroeconomic variables such as probability of recession and real GDP growth are important and along with state controls may further imply a role of government policy. Finally, the paper finds that occupation controls are important while industry controls do not appear to play a strong role.

Keywords: machine learning, restricted income profile, earnings instability, risk
JEL Codes: D8, J0, D3

1. Edit Introduction
2. Redo Theoretical Framework
3. Edit Data
4. Edit Empirical Strategy
5. Rewrite Results
6. Rewrite Conclusion
7. Consistent formatting of bibliography
8. Use old sections as well as suggested edits as an outline for rewritten sections
9. Lasso tables don't include ordering of FE groups

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1 Introduction

Understanding the nature of earnings risk is central to both individual decision making and policy. People are often thought to be risk averse not risk neutral and therefore earnings risk has an effect on how individuals make intertemporal decisions such as saving. Especially on the analysis side, structural life cycle models have to make assumptions about the nature of earnings dispersion and risk. Misspecifying the earnings process can lead to inaccurate or incorrect conclusions in many cases and this is why understanding the nature of earnings risk and the earnings process is so important.

To better pin down the quantitative importance of shocks versus profile heterogeneity, this paper builds on the work of (Drewianka and Oberg 2025, 2019) [5, 6] which uses a moment condition approach to test for heterogeneity in expected income processes. The paper uses a Restricted Income Profile (RIP) model to estimate lifetime earnings risk and then tests for heterogeneity in expected income processes using a moment condition approach.

Restricted Income Profile (RIP) models posit that all individuals with the same observed characteristics follow the same earnings trajectory. Deviations then being attributed to shocks to the individuals income process. In RIP, dispersion is then attributed to the variance of these shocks. (Abowd and Card, 1989; Meghir and Pistaferri, 2004) [1, 13] This means that the variance of shocks and variance in observed characteristics are sources of dispersion in earnings. Some of the variance is due to shocks the rest of the variance is due to the different expected income processes. (Guvenen 2007; Baker 1997) [7, 2]

This paper tests for predictors of lifetime earnings risk as parameterized by the model in Drewianka and Oberg (2025) [5]. The paper hopes to find which characteristics are important in explaining the variance of shocks in the RIP framework.

This paper finds that education and age are important in explaining lifetime earnings risk. The paper also finds that macroeconomic variables such as probability of recession and real GDP growth are important and play at least a small role and that the inclusion of state controls may further the hypothesis of government policy playing a role. Finally, the paper finds that occupation controls are important while industry controls do not appear to play a large role.

This paper contributes to multiple strands of literature. The first is the literature around (RIP) models and their use in life cycle models. (MaCurdy 1982; Abowd and Card 1989; Hryshko 2012). [12, 1, 10] In addition to that it also contributes to a second strand of literature focusing on lifetime earnings risk. (Drewianka and Oberg 2025; Guvenen 2007; Meghir and Pistaferri 2004). [5, 8, 13] Finally the paper contributes to the literature on machine learning and its use in economics by utilizing neural networks and SHAP values to

interpret the results (Lundberg and Lee, 2017). [11]

The rest of the paper is structured as follows. Section 2 discusses the model and data, Section 3 discusses the empirical strategy, Section 4 discusses the results, and Section 5 concludes the paper.

1. preview the results at the end
2. Also preview the methods used a little more
3. Spend a little more time explaining the contribution of this paper specifically, in both using some fancier methods and to identify covariance with this risk, but also the importance of being able to characterize people who are at an elevated risk in both permanent and transitory horizons
4. Do cross validation for RF and lasso?

2 Theoretical Framework

3 Data

The data used is the 1970-2023 waves of the Panel Study of Income Dynamics (PSID 2023). [14] The sample is of men between the ages of 22 and 69 with students being excluded. Many of the men in the sample come from the Survey of Economic Opportunity which has been shown to be representative of the main PSID sample (SEO; Hill 1992; Drewianka 2010). [9, 4] Income is calculated in 2015 dollars using the Consumer Price Index (CPI 2017). [3] The analysis uses several key variables, including education (categorized as less than high school, high school or some college, and bachelor's degree but less than a master's degree), age (with higher-order polynomial terms Age^2 and Age^3 to capture non-linear effects), and macroeconomic indicators such as the probability of a recession (P(Recession)) and real GDP growth. Individual-specific variables include fixed effects for wages, a moving average of income over the last five years (MA(Last 5 years income)), employment status (Employed), and job tenure (years in the current job). The primary outcome variables, gamma (γ) and alpha (α), measure lifetime earnings risk. The gamma statistic measures the variance in permanent income shocks while the alpha statistic measures the variance in temporary income shocks. Both these statistics are calculated using the moment condition approach outlined in Drewianka and Oberg (2025). [5]. The income variable used is the log of the individual's hourly wage. The hourly wage is calculated by dividing annual labor income by the product of weeks worked and usual hours worked per week.

Additionally, the analysis incorporates fixed effects for occupation, industry, state of residence, and other controls such as year, race, and cohort. With the theoretical framework and data laid out, the following section turns to the empirical implementation of the model.

1. Mention CNEF and how it was combined with the PSID
2. Mention that macro economic variables are from another source and cite it
3. Discuss inconsistencies with some of the variables across years, general PSID BS, tenure, occ, ind, race even
4. Add a table with summary stats maybe? Describing people with certain combinations of characteristics, possibly a concern for overfitting with ML methods

4 Empirical Strategy

The empirical strategy of this paper is broken into two parts. The first part is estimating gamma and alpha, then contruct a weighted gamma and alpha for each individual in each year since both statistics are across i, t, j, and q. (i - individual, t - time, j and q - period or "window" used in calculation). The second part of the empirical strategy is estimating the then weighted values of each statistic using various models and variables to look for trends in the characteristics of individuals in regards to lifetime earnings risk.

For the estimation of both gamma and alpha, consolidation across j and q is done by using a mixed regression for the gamma and then a fixed effects regression to get the composite or weighted value for each person year.

The following mixed regression is used to estimate gamma based on the above equation. This is done to get an estimate of gamma for each individual in each year across j and q. A mixed regression used instead of standard OLS because $\sigma^2_{\eta_i}$ is correlated with both the constant and the coefficient on (j+k+q) as seen in equation 8 and therefore a mixed regression is used to account for this by allowing the residuals to be separately correlated with (j+k+q) for each individual i in year t. (For more on this see Drewianka and Oberg (2025) [5]) The mixed regression is as follows:

$$\gamma_{itjq} = \beta_0 + \beta_1 (j + k + q) + \beta_2 j + \epsilon_{itjq}, \tag{1}$$

$$\epsilon_{itjq} = \beta_{0it} + \beta_{1it} (j + k + q) + e_{itjq}. \tag{2}$$

The random effects component (j+k+q) models how each person-year combination may have a unique relationship with the (j+k+q) variable while the cov(unstructured) option

in Stata allows for unrestricted correlation between the random intercept and slope. The standard errors are clustered at the person level to account for within-person correlation across time.

From here a regression is run across fixed effects for each combination of j and q with year effects absorbed. Weights are then calculated for each combination of j and q based on their accuracy in predicting gamma using inverse MSE. Specifically, after estimating the fixed effects model, the mean squared residual (MSE) is calculated for each (j,q) combination, then use the square root of the inverse of these values ($\sqrt{(1/MSE)}$) as weights. Then the weighted gamma is calculated consolidating across j and q so that there are only individual-year gamma values. Having obtained individual-year estimates of earnings risk, we now examine the determinants of this risk using several regression techniques.

The second part of the empirical strategy uses various models to estimate lifetime earnings risk (gamma). We begin with the simplest specification to establish a baseline before introducing more sophisticated selection and machine-learning methods. The first model is a standard OLS regression of the lifetime earnings risk variable, gamma, on various controls and variables. The second model is a stepwise regression which selects variables based on a p-value threshold. The third model is a lasso regression which penalizes the size of the coefficients to select variables. Finally, the fourth model uses a multi-layer perceptron and SHAP values to interpret the results.

The OLS regressions are fairly standard however the stepwise and lasso models do have components that are worth mentioning. For the stepwise regression model a cutoff p-value of 0.05 is used to select variables. The model removes the least significant variable (or group of variables in the case of a set of controls) in rounds until it reaches the cutoff. 0.05 was selected so that the model would still select some variables but the rankings of the many of the variables would be clear. A higher cutoff and most of the variables would be selected and cardinal rankings would not be visible. Too low of a cutoff value and many of the variables would not be selected at all.

As for the lasso model, the model is set up to select variables based on the Bayesian Information Criterion (BIC) which is a common method for selecting variables in penalized regression. While the model selects lambda based on cross-validation, the selected model isn't very relevant to the analysis as the rankings are so the selected model isn't discussed. It is worth mentioning however that the lasso model is a penalized regression and therefore some weight is assigned to the size of the coefficients when selecting variables. This is different from the stepwise regression which is indiscriminate regarding the size of the coefficient and only considered the significance of a variable.

Finally the multi-layer perceptron model is a neural network that is used to estimate

lifetime earnings risk. The model is trained on the same variables and data used by the OLS, stepwise, and lasso models. The model is 4 dense layers (excluding the input layer) with 1000 nodes each and a linear output layer at the end. The model used a sigmoid activation function in the hidden layers to allow for continuous support. This was done instead of a ReLU activation function to allow for more definition in the parameters instead of some parts of the perceptron being "dead" and not contributing and complicating the SHAP value interpretation. This model size and structure was selected as it achieved the best performance in terms of mean squared error (MSE). The model trained with MSE as its loss function and used early stopping to prevent overfitting and trained on 70 percent of the data with the rest being used for test and validation.

The SHAP values are then used to interpret the results of the multi-layer perceptron model. SHAP values are constructed by calculating marginal contribution of a variable as a deviation from the output variable's mean if the variable was changed. This is done across a sample of observations and gives each variable a distribution of SHAP values. The summary plot shows the distribution of SHAP values for the continuous variables.

1. be sure to mention Alpha and how alpha is estimated because it is estimated differently
2. explain some of the strengths and weaknesses of some of the models, particularly lasso
3. explain the neural network part much more in depth
4. explain the reasoning being behind using OLS, as in kitchen sink approach to just see some covariates, but also that it is subject to overfitting
5. explain the advantages and disadvantages of the step wise regression, particularly mention it's advantages and disadvantages in contrast to just standard OLS, along with similarities to OLS

5 Results

5.1 Permanent Shocks

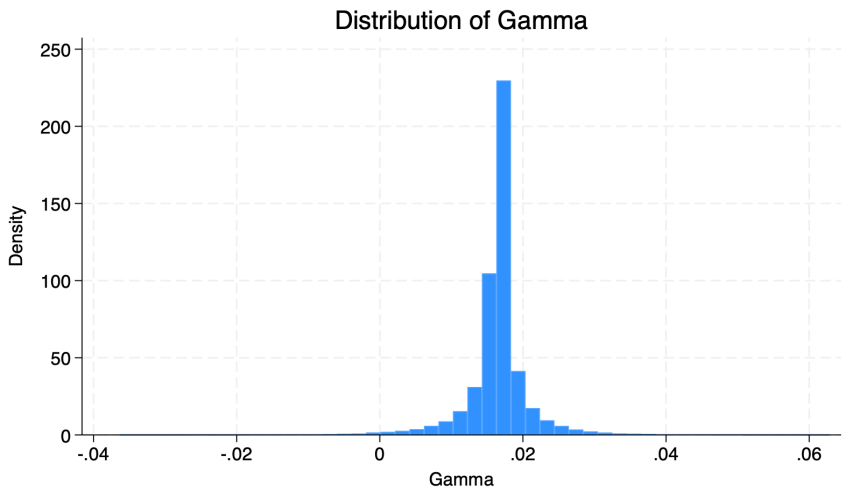

Figure 1: Distribution of Gamma

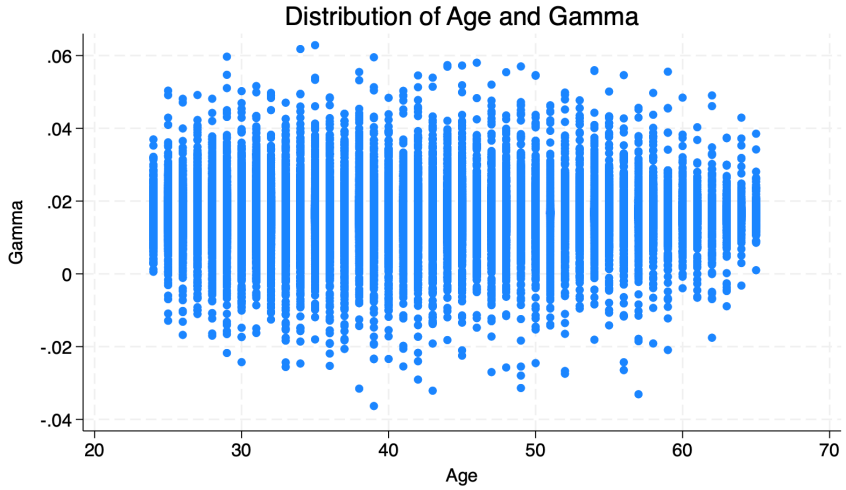

Figure 2: Scatterplot of Age vs. Gamma

Table 1: Gamma Regressions: F-Test Results

	No Occ/Ind	No Occ	No Ind	All Controls
	(1)	(2)	(3)	(4)
State FE	1.13 (0.2448)	1.11 (0.2704)	1.16 (0.2082)	1.12 (0.2601)
Year FE	0.99 (0.4753)	1.20 (0.2418)	1.20 (0.2353)	1.22 (0.2245)
Race FE	0.74 (0.5919)	0.70 (0.6206)	0.67 (0.6500)	0.71 (0.6172)
Cohort FE	5.02*** (0.0018)	4.78*** (0.0025)	4.57*** (0.0033)	4.71*** (0.0027)
Occupation FE	–	–	1.37** (0.0180)	1.48*** (0.0038)
Industry FE	–	0.68 (0.9147)	–	0.96 (0.5326)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 2: Gamma Regressions: Industry Selection Results

Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Constr. Relate	3	1	1
Other Services	24	2	4
Mechanical Eng	1	3	8
Not Applicable	6	4	2
Other Trans.	18	5	6
Wholesale	19	6	13
Iron/Steel	33	7	19
Legal Services	7	8	5
Wood/Paper/Print	4	9	16
Construction	13	10	9
Social Security	23	11	25
Clothing/Text	21	12	23
Insurance	5	13	28
Retail	20	14	3
Chemicals	30	15	24
Electrical Eng	8	16	27
Restaurants	11	17	29
Health Service	9	18	17
Synthetics	32	19	21
Energy/Water	12	20	18
Train System	16	21	31
Agric.,Forestry	27	22	7
Volunt./Church	28	23	26
Food Industry	22	24	10
Educ./Sport	25	25	20
Public Administration	29	26	11
Service Indust	17	27	12
Earth/Clay/Stone	26	28	32
Mining	10	29	30
Postal System	31	30	14
Fisheries	15	31	22
Financial Inst	2	32	15
Priv. Househld	14	33	33

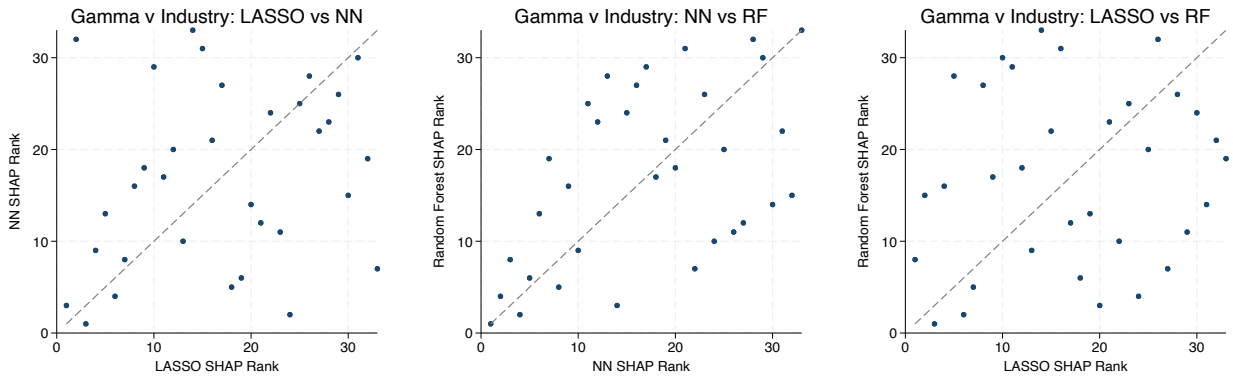

Figure 3: Gamma Industry SHAP Rank Comparisons

Table 3: Gamma Regressions: Occupation Selection Results

Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Priv.Bus.Leadr	67	1	1
Bricklay/Carpt	27	2	11
Pipe Fitter	3	3	2
Transport.Oper	26	4	12
Electr. Enginr	13	5	10
Farm Manager	15	6	4
Mailman	42	7	31
Insurance Rep.	2	8	5
Soldier	45	9	6
Hair Stylist	44	10	16
Vendor	34	11	14
Tailor	69	12	9
Inspector	46	13	39
Labor/Craftsmn	77	14	29
RelatMedicalJob	64	15	37
Ofc.Worker Etc	4	16	38
Educator	56	17	18
Architect/Engineer	65	18	17
Dr./Dentist/Vet	40	19	24
Sculptr/Paintr	11	20	27
Legislator	41	21	32
Prof. Athlete	51	22	52
Bookkeepr/Cash	48	23	64
SecurityServic	47	24	20
Tool/Die Maker	54	25	48
Farm Hand	38	26	61
Painter	39	27	26
Cabinet Maker	63	28	59
Agriculturladm	68	29	44
Lumbrman/Axman	60	30	60
BusinessManagr	8	31	22
Cleric	50	32	13
Eng.Tech.Expert	59	33	40
Author	9	34	47
Janitor	49	35	42
Scientist	58	36	28
Foundry Worker	35	37	55
ComputerOperat	32	38	63
Administrator	23	39	45
Stat.Mach.Oper	25	40	19
Domestic Help	5	41	76
Machine Fitter	21	42	15
Music/Perform	53	43	41
Cook/Waiter	33	44	30
Dry-Cleaner	6	45	21
Broadcaster	72	46	53
Life/Phys. Sci.	61	47	57
Rest./StoreMgr	75	48	7
Jewelry Maker	37	49	77
Shoemaker	73	50	65
Transp. Attend	22	51	51
Not Applicable	1	52	3
ChemicalWorker	70	53	68
Food Producer	7	54	35
Aero/MarineEngr	20	55	33
Convey. Oper.	12	56	8
Mathematician	16	57	50
Economist	62	58	25
Conductor	74	59	36
Office Manager	36	60	62
Salesperson	28	61	56
Fisher/Hunter	18	62	46
Accountant	19	63	54
Tech.Salespers	31	64	43
Lawyer	14	65	23
Stenographer	24	66	67
Chemist	17	67	69
Glazier	57	68	75
Forestry Work	71	69	71
Manufacturer	30	70	73
Service Worker	66	71	49
Spinner/Weaver	43	72	70
Buyer	10	73	58
Miner	55	74	66
Printer Etc.	52	75	34
Stone Cutter	76	76	74
Tel. Operator	29	77	72
HH Supervisor			

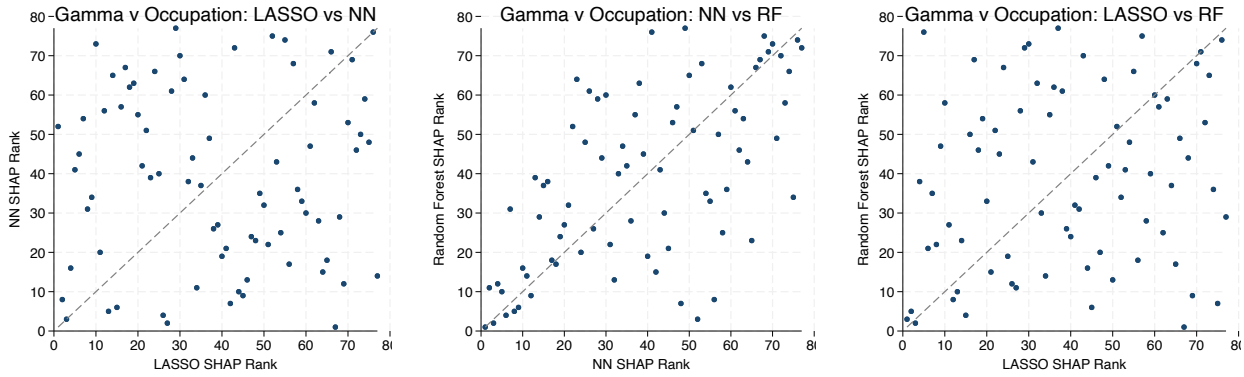

Figure 4: Gamma Occupation SHAP Rank Comparisons

Table 4: Gamma Regressions: OLS Results (Coefficients x 100)

	(1)	(2)	(3)	(4)	(5)
	Gamma	Gamma	Gamma	Gamma	Gamma
Less than High School	-0.00236	0.000451	0.00139	-0.00193	-0.0000625
	(0.00573)	(0.00619)	(0.00651)	(0.00686)	(0.00692)
High School Graduate	-0.00378	-0.00316	-0.00310	-0.00468	-0.00338
	(0.00427)	(0.00441)	(0.00469)	(0.00507)	(0.00513)
Some College	-0.0112**	-0.0113**	-0.0110**	-0.0115**	-0.0111*
	(0.00537)	(0.00542)	(0.00560)	(0.00578)	(0.00582)
Probability of Recession	0.00000979	0.0578	-0.00417	-0.00414	-0.00452
	(0.000127)	(0.0517)	(277.6)	(277.5)	(277.5)
Real GDP growth rate	0.000460	0.0208**	0.0152	0.0151	0.0165
	(0.00116)	(0.00939)	(21.45)	(21.45)	(21.45)
5-year moving average of AEP	-0.000193**	-0.000221**	-0.000168*	-0.000232**	-0.000205**
	(0.0000855)	(0.0000933)	(0.0000987)	(0.000101)	(0.000103)
Out of Labor Force	0.0138	0.0156	0.0199	0.0183	0.0201
	(0.0154)	(0.0155)	(0.0159)	(0.0159)	(0.0159)
Tenure	-0.0000347	0.0000451	-0.0000164	0.0000808	0.0000686
	(0.000256)	(0.000278)	(0.000285)	(0.000286)	(0.000287)
Age (cubic)	Yes	Yes	Yes	Yes	Yes
State FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes
R-squared	0.000	0.002	0.002	0.003	0.003
N	82357	82333	81556	81556	81556

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

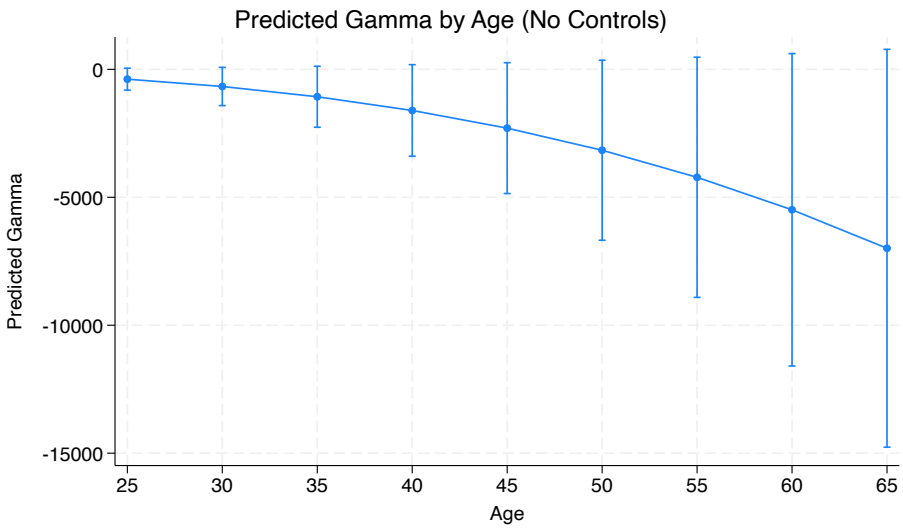

Figure 5: Predicted Gamma by Age (No Controls)

Table 5: Gamma Regressions: Stepwise Selection Results (Coefficients x 100)

	(1)	(2)	(3)	(4)	(5)
	Gamma	Gamma	Gamma	Gamma	Gamma
Age	-0.00370**	-0.0186*	-0.0190*	-0.0192*	-0.0192*
	(0.00135)	(0.00778)	(0.00784)	(0.00786)	(0.00786)
5-year moving average of AEP	-0.000201**	-0.000210**	-0.000204**	-0.000239**	-0.000239**
	(0.0000757)	(0.0000768)	(0.0000776)	(0.0000893)	(0.0000893)
Some College	-0.00913*	-0.00999*	-0.00964*		
	(0.00462)	(0.00464)	(0.00468)		
Age Squared	0.0000429**	0.000415*	0.000424*	0.000427*	0.000427*
	(0.0000160)	(0.000188)	(0.000190)	(0.000191)	(0.000191)
Age Cubed		-0.00000291*	-0.00000299*	-0.00000299*	-0.00000299*
		(0.00000147)	(0.00000149)	(0.00000149)	(0.00000149)
Constant	0.0173***	0.0192***	0.0192***	0.0195***	0.0195***
	(0.000280)	(0.00104)	(0.00104)	(0.00106)	(0.00106)
State FE					
Year FE					
Race FE					
Cohort FE		✓	✓	✓	✓
Occupation FE				✓	✓
Industry FE					
State FE Available	No	Yes	Yes	Yes	Yes
Year FE Available	No	Yes	Yes	Yes	Yes
Race FE Available	No	Yes	Yes	Yes	Yes
Cohort FE Available	No	Yes	Yes	Yes	Yes
Occupation FE Available	No	No	No	Yes	Yes
Industry FE Available	No	No	Yes	No	Yes
R-squared	0.000	0.000	0.000	0.002	0.002
N	82357	82333	81556	81556	81556

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 6: Gamma Regressions: Lasso Selection Results

	No Controls	No Occ/Ind	No Occ	No Ind	All Controls
EDU1	6	9	7	8	8
EDU2	4	6	6	5	6
EDU3	1	2	2	2	2
OLF	2	5	4	4	5
PrRecess	8				
currentage	4	7	8	7	7
currentagecube	5	4	3	2	3
currentagesq	9	10	10	9	9
ma5aep	1	3	5	3	4
rGDPgrow	3				
tenure	7	8	9	6	7
State FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes

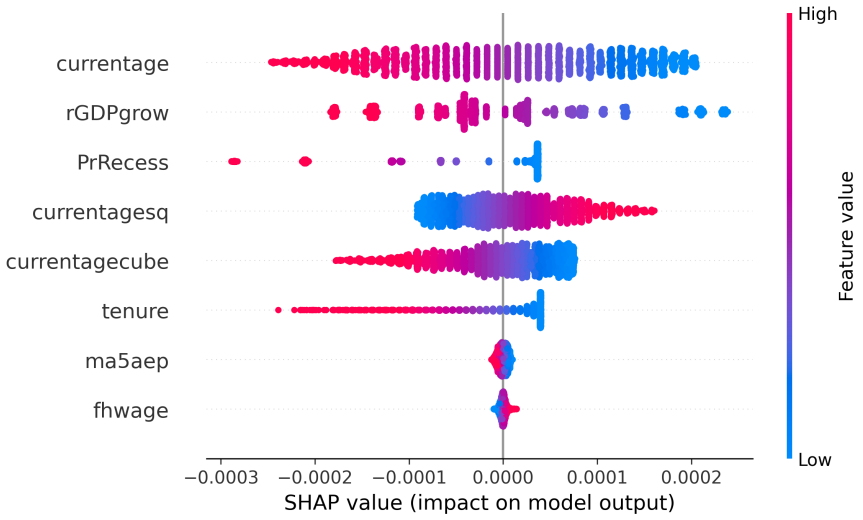

Figure 6: SHAP Summary Plot for Gamma

5.2 Transitory Shocks

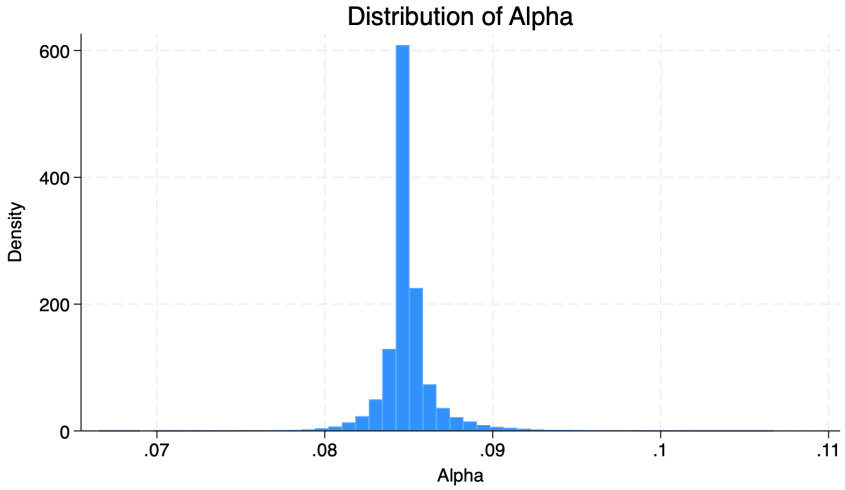

Figure 7: Distribution of Alpha

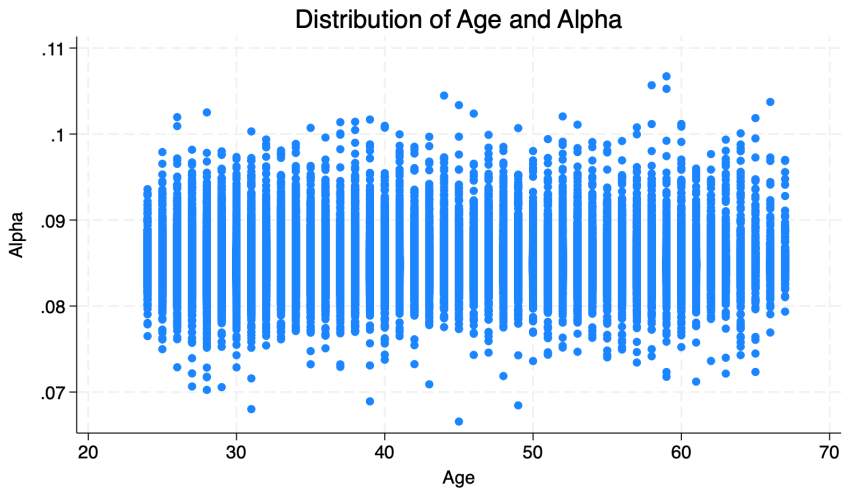

Figure 8: Scatterplot of Age vs. Alpha

Table 7: Alpha Regressions: F-Test Results

	No Occ/Ind	No Occ	No Ind	All Controls
	(1)	(2)	(3)	(4)
State FE	3.33*** (0.0000)	3.01*** (0.0000)	3.17*** (0.0000)	2.99*** (0.0000)
Year FE	2.94*** (0.0000)	2.92*** (0.0000)	3.08*** (0.0000)	2.98*** (0.0000)
Race FE	2.83** (0.0148)	3.52*** (0.0035)	3.15*** (0.0076)	3.63*** (0.0027)
Cohort FE	10.62*** (0.0000)	10.79*** (0.0000)	10.82*** (0.0000)	11.16*** (0.0000)
Occupation FE	–	–	3.86*** (0.0000)	3.20*** (0.0000)
Industry FE	–	6.14*** (0.0000)	–	4.55*** (0.0000)

Note: F-statistics reported with p-values in parentheses.

* p<0.10, ** p<0.05, *** p<0.01

Table 8: Alpha Regressions: Lasso Industry Selection Results

Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Mechanical Eng	16	1	8
Agric.,Forestry	10	2	13
Public Administration	19	3	9
Wood/Paper/Print	33	4	15
Retail	31	5	3
Wholesale	13	6	12
Iron/Steel	28	7	18
Health Service	17	8	20
Educ./Sport	23	9	14
Postal System	9	10	19
Other Trans.	8	11	10
Financial Inst	26	12	11
Construction	3	13	17
Electrical Eng	6	14	26
Energy/Water	22	15	7
Clothing/Text	32	16	21
Food Industry	14	17	29
Insurance	11	18	16
Synthetics	2	19	22
Chemicals	5	20	32
Volunt./Church	15	21	28
Other Services	12	22	4
Legal Services	27	23	5
Train System	25	24	30
Mining	30	25	24
Earth/Clay/Stone	20	26	27
Service Indust	18	27	6
Restaurants	4	28	23
Fisheries	7	29	25
Social Security	21	30	31
Constr. Relate	1	31	1
Priv. Househld	24	32	33
Not Applicable	29		2

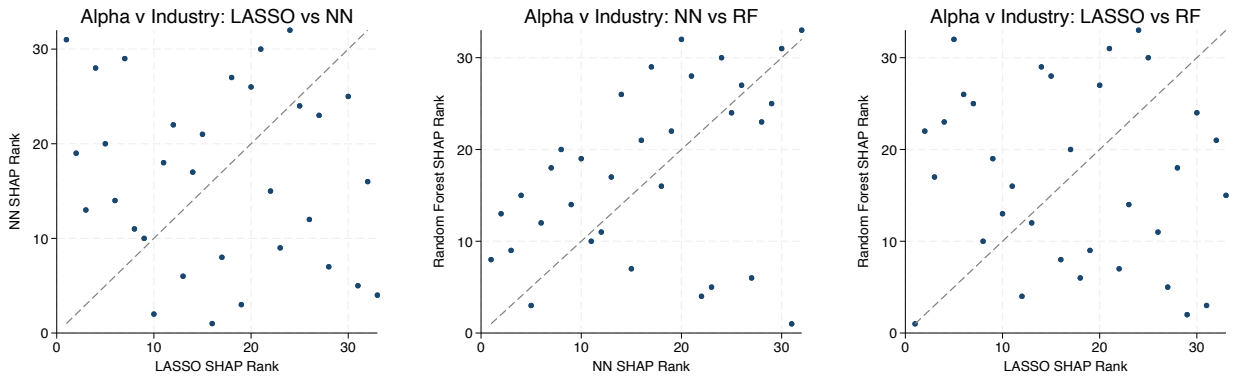

Figure 9: Alpha Industry SHAP Rank Comparisons

Table 9: Alpha Regressions: Lasso Occupation Selection Results

Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Farm Manager	13	1	9
Labor/Craftsmn	23	2	5
Machine Fitter	36	3	13
Mathematician	51	4	19
Mailman	70	5	32
Tool/Die Maker	37	6	25
Convey. Oper.	25	7	8
Architect/Engineer	57	8	10
Inspector	9	9	37
BusinessManagr	67	10	21
Ofc.Worker Etc	68	11	14
Vendor	63	12	12
Priv.Bus.Leadr	60	13	2
Transport.Oper	24	14	1
Eng.Tech.Expert	56	15	29
Educator	47	16	4
Electr. Enginr	35	17	16
Insurance Rep.	64	18	6
Accountant	41	19	35
Janitor	38	20	11
Scientist	58	21	24
Food Producer	3	22	38
SecurityServic	16	23	23
Life/Phys. Sci.	54	24	42
Printer Etc.	30	25	58
Pipe Fitter	33	26	7
RelatMedicalJob	52	27	43
Cook/Waiter	39	28	48
Soldier	77	29	15
Stat.Mach.Oper	26	30	53
Service Worker	15	31	63
Office Manager	61	32	65
Cleric	46	33	55
Painter	29	34	52
Bookkeepr/Cash	74	35	34
Administrator	76	36	56
Dr./Dentist/Vet	53	37	18
Aero/MarineEngr	55	38	31
Music/Perform	43	39	30
ComputerOperat	73	40	50
Bricklay/Carpt	27	41	17
Fisher/Hunter	2	42	39
Foundry Worker	7	43	33
Agriculturladm	14	44	45
Lawyer	48	45	26
ChemicalWorker	5	46	62
Spinner/Weaver	4	47	72
Chemist	40	48	44
Miner	8	49	22
Buyer	66	50	40
Salesperson	62	51	47
Rest./StoreMgr	50	52	27
Domestic Help	11	53	69
Author	45	54	36
Prof. Athlete	42	55	68
Tailor	19	56	28
Cabinet Maker	21	57	66
Sculptr/Paintr	44	58	41
Tech.Salespers	65	59	54
Legislator	59	60	20
Dry-Cleaner	18	61	60
Stenographer	75	62	61
Glazier	31	63	77
Shoemaker	20	64	67
Farm Hand	12	65	70
Jewelry Maker	32	66	71
Conductor	71	67	59
Forestry Work	10	68	76
Lumbrman/Axman	6	69	49
Broadcaster	34	70	64
Manufacturer	28	71	74
Tel. Operator	69	72	73
Transp. Attend	72	73	46
Hair Stylist	17	74	51
Economist	49	75	57
HH Supervisor		76	
Stone Cutter	22	77	75
Not Applicable	1		3

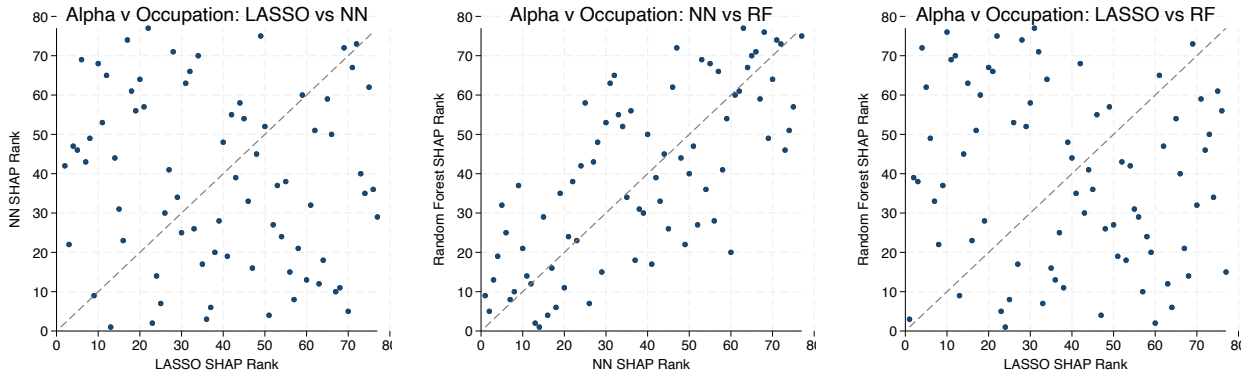

Figure 10: Alpha Occupation SHAP Rank Comparisons

Table 10: Alpha Regressions: OLS Results (Coefficients x 100)

	(1)	(2)	(3)	(4)	(5)
	Alpha	Alpha	Alpha	Alpha	Alpha
Less than High School	0.00368**	0.00436**	0.00249	0.00391*	0.00316
	(0.00185)	(0.00200)	(0.00211)	(0.00223)	(0.00224)
High School Graduate	0.00548***	0.00604***	0.00501***	0.00613***	0.00551***
	(0.00141)	(0.00145)	(0.00155)	(0.00168)	(0.00169)
Some College	0.00777***	0.00762***	0.00683***	0.00692***	0.00651***
	(0.00177)	(0.00178)	(0.00186)	(0.00192)	(0.00193)
Probability of Recession	0.0000674	0.00135	0.0127	0.0128	0.0135
	(0.0000411)	(76.56)	(0.0322)	(0.0322)	(0.0322)
Real GDP growth rate	0.000168	-0.00696	-0.00785**	-0.00784**	-0.00809**
	(0.000378)	(0.666)	(0.00317)	(0.00316)	(0.00317)
5-year moving average of AEP	0.000172***	0.000149***	0.0000996***	0.000155***	0.000129***
	(0.0000273)	(0.0000297)	(0.0000318)	(0.0000323)	(0.0000332)
Out of Labor Force	0.0113**	0.0107**	0.00955**	0.00991**	0.00932**
	(0.00447)	(0.00448)	(0.00468)	(0.00467)	(0.00468)
Tenure	-0.000374***	-0.000522***	-0.000565***	-0.000576***	-0.000572***
	(0.0000818)	(0.0000892)	(0.0000922)	(0.0000925)	(0.0000928)
Age (cubic)	Yes	Yes	Yes	Yes	Yes
State FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes
R-squared	0.002	0.005	0.007	0.008	0.009
N	102946	102910	100781	100781	100781

Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

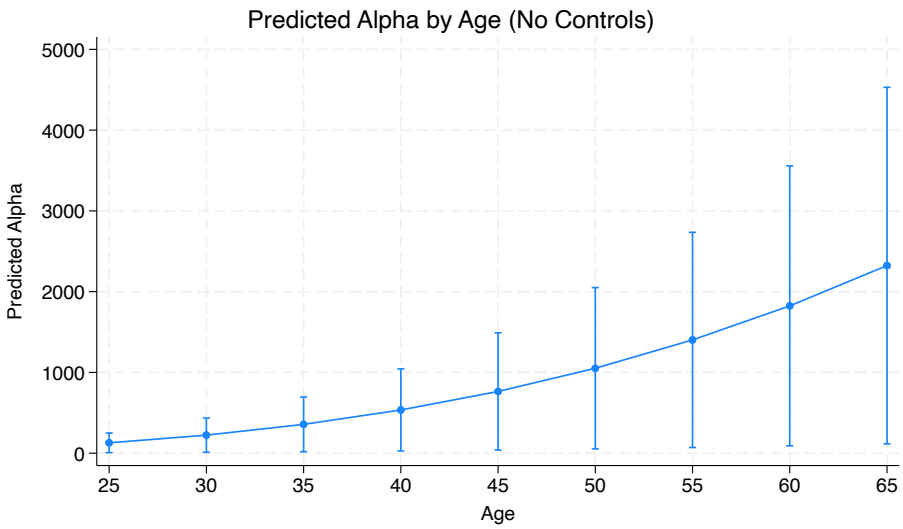

Figure 11: Predicted Alpha by Age (No Controls)

Table 11: Alpha Regressions: Stepwise Selection Results (Coefficients x 100)

	(1) Alpha	(2) Alpha	(3) Alpha	(4) Alpha	(5) Alpha
Less than High School	0.00362*	0.00421*			
	(0.00182)	(0.00199)			
High School Graduate	0.00542***	0.00594***	0.00386**	0.00418**	0.00389**
	(0.00140)	(0.00144)	(0.00123)	(0.00128)	(0.00128)
Some College	0.00776***	0.00759***	0.00595***	0.00546**	0.00532**
	(0.00177)	(0.00178)	(0.00169)	(0.00172)	(0.00173)
Probability of Recession	0.0000525*				
	(0.0000259)				
Age Cubed	7.66e-08***	0.00000111**	0.00000106*	0.00000114**	0.00000110*
	(9.43e-09)	(0.000000421)	(0.000000429)	(0.000000429)	(0.000000429)
5-year moving average of AEP	0.000170***	0.000151***	0.000113***	0.000172***	0.000142***
	(0.0000262)	(0.0000296)	(0.0000295)	(0.0000311)	(0.0000320)
Out of Labor Force	0.0121**	0.0107*	0.00942*	0.00972*	0.00919*
	(0.00445)	(0.00448)	(0.00468)	(0.00467)	(0.00468)
Tenure	-0.000379***	-0.000512***	-0.000541***	-0.000553***	-0.000543***
	(0.0000816)	(0.0000881)	(0.0000903)	(0.0000907)	(0.0000909)
Age		0.00540*	0.00505*	0.00538*	0.00518*
		(0.00232)	(0.00236)	(0.00236)	(0.00236)
Age Squared		-0.000136*	-0.000129*	-0.000138*	-0.000133*
		(0.0000550)	(0.0000560)	(0.0000560)	(0.0000560)
Constant	0.0848***	0.0842***	0.0842***	0.0841***	0.0841***
	(0.0000155)	(0.000366)	(0.000373)	(0.000374)	(0.000374)
State FE		✓	✓	✓	✓
Year FE		✓	✓	✓	✓
Race FE		✓	✓	✓	✓
Cohort FE		✓	✓	✓	✓
Occupation FE				✓	✓
Industry FE			✓		✓
State FE Available	No	Yes	Yes	Yes	Yes
Year FE Available	No	Yes	Yes	Yes	Yes
Race FE Available	No	Yes	Yes	Yes	Yes
Cohort FE Available	No	Yes	Yes	Yes	Yes
Occupation FE Available	No	No	No	Yes	Yes
Industry FE Available	No	No	Yes	No	Yes
R-squared	0.002	0.005	0.007	0.008	0.009
N	102946	102910	100781	100781	100781

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 12: Alpha Regressions: Lasso Selection Results

	No Controls	No Occ/Ind	No Occ	No Ind	All Controls
EDU1	8	7	8	7	7
EDU2	6	5	6	5	5
EDU3	5	5	5	5	5
OLF	3	4	4	4	4
PrRecess	7				
currentage		8	9	8	8
currentagecube	2	6	7	6	6
currentagesq					
ma5aep	1	2	3	3	3
rGDPgrow	9				
tenure	4	3	2	2	2
State FE	No	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes	Yes
Race FE	No	Yes	Yes	Yes	Yes
Cohort FE	No	Yes	Yes	Yes	Yes
Occupation FE	No	No	No	Yes	Yes
Industry FE	No	No	Yes	No	Yes

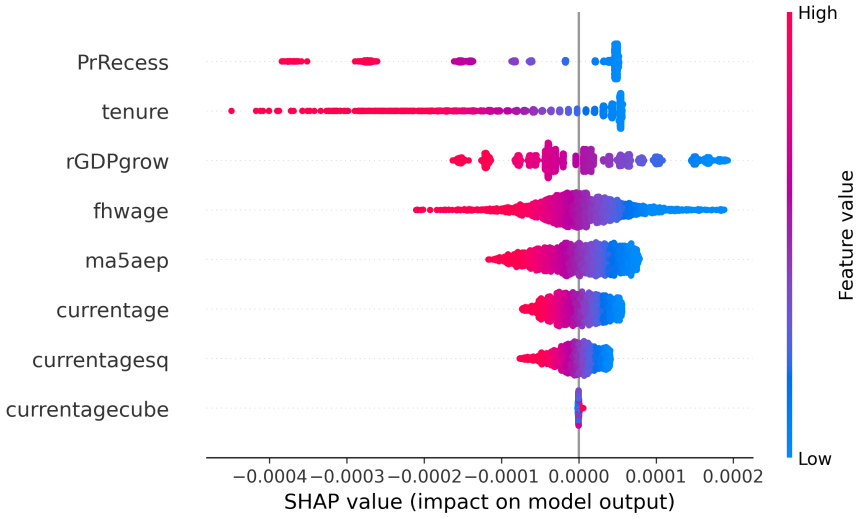

Figure 12: SHAP Summary Plot for Alpha

6 Conclusion

References

[1] Abowd, J. M. and Card, D. (1989). On the Covariance Structure of Earnings and Hours Changes. *Econometrica*, 57(2), 411–445.

[2] Baker, M. (1997). Growth-Rate Heterogeneity and the Covariance Structure of Life-Cycle Earnings. *Journal of Labor Economics*, 15(2), 338–375.

[3] Consumer Price Index for All Urban Consumers (2017). Data set, series CUUR0000SA0. Washington, D.C.: U.S. Bureau of Labor Statistics.

[4] Drewianka, S. (2010). Cross-Sectional Variation in Individuals' Earnings Instability. *Review of Income and Wealth*, 56(2), 291–326.

[5] Drewianka, Scott and Oberg, Phillip. (2025). Earnings Risk and Heterogeneous Expected Earnings Profiles. Quantitative Economics. Forthcoming.

[6] Drewianka, S. and Oberg, P. (2019). Estimating Heterogeneity in Lifetime Earnings Risk. Unpublished manuscript, University of Wisconsin–Milwaukee and Illinois Wesleyan University. Available at `http://dx.doi.org/10.2139/ssrn.3630477`.

[7] Guvenen, F. (2009). An Empirical Investigation of Labor Income Processes. *Review of Economic Dynamics*, 12, 58–79.

[8] Guvenen, F. (2007). Learning Your Earning: Are Labor Income Shocks Really Very Persistent? *American Economic Review*, 97(3), 687–712.

[9] Hill, M. S. (1992). *The Panel Study of Income Dynamics: A User's Guide*, Volume 2. Newbury Park, CA: Sage Publications.

[10] Hryshko, D. (2012). Labor Income Profiles Are Not Heterogeneous: Evidence from Income Growth Rates. *Quantitative Economics*, 3, 177–209.

[11] Lundberg, S. M., and Lee, S. I. (2017). A unified approach to interpreting model predictions. Advances in neural information processing systems, 30.

[12] MaCurdy, T. E. (1982). The Use of Time Series Processes to Model the Error Structure of Earnings in a Longitudinal Data Analysis. *Journal of Econometrics*, 18(1), 83–114.

[13] Meghir, C. and Pistaferri, L. (2004). Income Variance Dynamics and Heterogeneity. *Econometrica*, 72(1), 1–32.

[14] Panel Study of Income Dynamics (2023). Public use dataset. Ann Arbor, MI: Institute for Social Research, Survey Research Center, University of Michigan.

A Appendix

Table 13: Gamma Regressions: Lasso Industry Selection Results

Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Constr. Relate	3	1	1
Other Services	24	2	4
Mechanical Eng	1	3	8
Not Applicable	6	4	2
Other Trans.	18	5	6
Wholesale	19	6	13
Iron/Steel	33	7	19
Legal Services	7	8	5
Wood/Paper/Print	4	9	16
Construction	13	10	9
Social Security	23	11	25
Clothing/Text	21	12	23
Insurance	5	13	28
Retail	20	14	3
Chemicals	30	15	24
Electrical Eng	8	16	27
Restaurants	11	17	29
Health Service	9	18	17
Synthetics	32	19	21
Energy/Water	12	20	18
Train System	16	21	31
Agric.,Forestry	27	22	7
Volunt./Church	28	23	26
Food Industry	22	24	10
Educ./Sport	25	25	20
Public Administration	29	26	11
Service Indust	17	27	12
Earth/Clay/Stone	26	28	32
Mining	10	29	30
Postal System	31	30	14
Fisheries	15	31	22
Financial Inst	2	32	15
Priv. Househld	14	33	33

Table 14: Gamma Regressions: Lasso Occupation Selection Results

Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Priv.Bus.Leadr	67	1	1
Bricklay/Carpt	27	2	11
Pipe Fitter	3	3	2
Transport.Oper	26	4	12
Electr. Enginr	13	5	10
Farm Manager	15	6	4
Mailman	42	7	31
Insurance Rep.	2	8	5
Soldier	45	9	6
Hair Stylist	44	10	16
Vendor	34	11	14
Tailor	69	12	9
Inspector	46	13	39
Labor/Craftsmn	77	14	29
RelatMedicalJob	64	15	37
Ofc.Worker Etc	4	16	38
Educator	56	17	18
Architect/Engineer	65	18	17
Dr./Dentist/Vet	40	19	24
Sculptr/Paintr	11	20	27
Legislator	41	21	32
Prof. Athlete	51	22	52
Bookkeepr/Cash	48	23	64
SecurityServic	47	24	20
Tool/Die Maker	54	25	48
Farm Hand	38	26	61
Painter	39	27	26
Cabinet Maker	63	28	59
Agriculturladm	68	29	44
Lumbrman/Axman	60	30	60
BusinessManagr	8	31	22
Cleric	50	32	13
Eng.Tech.Expert	59	33	40
Author	9	34	47
Janitor	49	35	42
Scientist	58	36	28
Foundry Worker	35	37	55
ComputerOperat	32	38	63
Administrator	23	39	45
Stat.Mach.Oper	25	40	19
Domestic Help	5	41	76
Machine Fitter	21	42	15
Music/Perform	53	43	41
Cook/Waiter	33	44	30
Dry-Cleaner	6	45	21
Broadcaster	72	46	53
Life/Phys. Sci.	61	47	57
Rest./StoreMgr	75	48	7
Jewelry Maker	37	49	77
Shoemaker	73	50	65
Transp. Attend	22	51	51
Not Applicable	1	52	3
ChemicalWorker	70	53	68
Food Producer	7	54	35
Aero/MarineEngr	20	55	33
Convey. Oper.	12	56	8
Mathematician	16	57	50
Economist	62	58	25
Conductor	74	59	36
Office Manager	36	60	62
Salesperson	28	61	56
Fisher/Hunter	18	62	46
Accountant	19	63	54
Tech.Salespers	31	64	43
Lawyer	14	65	23
Stenographer	24	66	67
Chemist	17	67	69
Glazier	57	68	75
Forestry Work	71	69	71
Manufacturer	30	70	73
Service Worker	66	71	49
Spinner/Weaver	43	72	70
Buyer	10	73	58
Miner	55	74	66
Printer Etc.	52	75	34
Stone Cutter	76	76	74
Tel. Operator	29	77	72
HH Supervisor			

Table 15: Alpha Regressions: Lasso Industry Selection Results

Industry	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Mechanical Eng	16	1	8
Agric.,Forestry	10	2	13
Public Administration	19	3	9
Wood/Paper/Print	33	4	15
Retail	31	5	3
Wholesale	13	6	12
Iron/Steel	28	7	18
Health Service	17	8	20
Educ./Sport	23	9	14
Postal System	9	10	19
Other Trans.	8	11	10
Financial Inst	26	12	11
Construction	3	13	17
Electrical Eng	6	14	26
Energy/Water	22	15	7
Clothing/Text	32	16	21
Food Industry	14	17	29
Insurance	11	18	16
Synthetics	2	19	22
Chemicals	5	20	32
Volunt./Church	15	21	28
Other Services	12	22	4
Legal Services	27	23	5
Train System	25	24	30
Mining	30	25	24
Earth/Clay/Stone	20	26	27
Service Indust	18	27	6
Restaurants	4	28	23
Fisheries	7	29	25
Social Security	21	30	31
Constr. Relate	1	31	1
Priv. Househld	24	32	33
Not Applicable	29		2

Table 16: Alpha Regressions: Lasso Occupation Selection Results

Occupation	LASSO SHAP Order	NN SHAP Order	RF SHAP Order
Farm Manager	13	1	9
Labor/Craftsmn	23	2	5
Machine Fitter	36	3	13
Mathematician	51	4	19
Mailman	70	5	32
Tool/Die Maker	37	6	25
Convey. Oper.	25	7	8
Architect/Engineer	57	8	10
Inspector	9	9	37
BusinessManagr	67	10	21
Ofc.Worker Etc	68	11	14
Vendor	63	12	12
Priv.Bus.Leadr	60	13	2
Transport.Oper	24	14	1
Eng.Tech.Expert	56	15	29
Educator	47	16	4
Electr. Enginr	35	17	16
Insurance Rep.	64	18	6
Accountant	41	19	35
Janitor	38	20	11
Scientist	58	21	24
Food Producer	3	22	38
SecurityServic	16	23	23
Life/Phys. Sci.	54	24	42
Printer Etc.	30	25	58
Pipe Fitter	33	26	7
RelatMedicalJob	52	27	43
Cook/Waiter	39	28	48
Soldier	77	29	15
Stat.Mach.Oper	26	30	53
Service Worker	15	31	63
Office Manager	61	32	65
Cleric	46	33	55
Painter	29	34	52
Bookkeepr/Cash	74	35	34
Administrator	76	36	56
Dr./Dentist/Vet	53	37	18
Aero/MarineEngr	55	38	31
Music/Perform	43	39	30
ComputerOperat	73	40	50
Bricklay/Carpt	27	41	17
Fisher/Hunter	2	42	39
Foundry Worker	7	43	33
Agriculturladm	14	44	45
Lawyer	48	45	26
ChemicalWorker	5	46	62
Spinner/Weaver	4	47	72
Chemist	40	48	44
Miner	8	49	22
Buyer	66	50	40
Salesperson	62	51	47
Rest./StoreMgr	50	52	27
Domestic Help	11	53	69
Author	45	54	36
Prof. Athlete	42	55	68
Tailor	19	56	28
Cabinet Maker	21	57	66
Sculptr/Paintr	44	58	41
Tech.Salespers	65	59	54
Legislator	59	60	20
Dry-Cleaner	18	61	60
Stenographer	75	62	61
Glazier	31	63	77
Shoemaker	20	64	67
Farm Hand	12	65	70
Jewelry Maker	32	66	71
Conductor	71	67	59
Forestry Work	10	68	76
Lumbrman/Axman	6	69	49
Broadcaster	34	70	64
Manufacturer	28	71	74
Tel. Operator	69	72	73
Transp. Attend	72	73	46
Hair Stylist	17	74	51
Economist	49	75	57
HH Supervisor		76	
Stone Cutter	22	77	75
Not Applicable	1		3